Perfectization and perfectoid crystals.

Setup

In this talk:

1) $\mathfrak{X}$ is a bounded$^{(*)}$ p-adic formal scheme. This means that $\mathfrak{X} = \bigcup \mathrm{Spf}(R)$ in the Zar topology where R is a p-complete and bounded p^∞-torsion

"p-adic formal scheme" will mean "bounded" p-adic formal.

(*) This means that $\exists n \in \mathbb{N}$ s.t if $p^m r = 0$ in R for some $m \gg 0$ then $p^n r = 0$. (Automatic if R Noetherian)

The main benefit from boundedness is that derived p-completion $=$ classical p-completion.

2) $\mathrm{SPfd}^{\mathrm{nil}}$ is the category of p-nilpotent semiperfectoid rings.

It is a basis of p-nilpotent rings for the flat topology.

Every object will be seen through the lens of it's functor of points on $\mathrm{SPfd}^{\mathrm{nil}}$

e.g a) S p-nilpotent:

$$\mathrm{Spec}(S) : \mathrm{SPfd}^{\mathrm{nil}} \longrightarrow \mathrm{Set}$$
$$A \longmapsto \mathrm{Ring}(A, S)$$

b) R p-complete

$$\mathrm{Spf}(S) := \varinjlim_n \mathrm{Spec}(R/p^n R)$$

(in (pre)sheaves)

1) Untilts and perfectization

Def $S \in \mathrm{SPfd}^{nil}$. An untilt of S is

$$S^\# \longrightarrow S$$

$S^\#$ perfectoid, with $(S^\#)^\flat \xrightarrow{\sim} S^\flat$.

$\square$

Remark: Automatically $S^\# \twoheadrightarrow S$. (exercise)

Remark: Equivalently, choice of (perfect) prism structure $(W(S^\flat), I)$ s.t. $I \subseteq \ker(\theta_S : W(S^\flat) \longrightarrow S)$ (exercise)

Def For any presheaf (of anima) $\mathcal{X}$ on SPfd^{nil} define the presheaf

$$\mathcal{X}^{pfd}(S) = \{ \operatorname{Spec}(S) \underset{\text{untilt}}{\to} \operatorname{Spf}(S^\#) \underset{\text{a map}}{\to} \mathcal{X} \}$$

Properties. 0) There is a natural map $\pi : \mathcal{X}^{pfd} \longrightarrow \mathcal{X}$. $\square$

1) Discreteness

Automorphisms of untilts are $\{id\}$

If

$$\begin{tikzcd} S^\# \arrow[rd, "f"] \arrow[d, "\sigma"', "\wr"] & \\ S^\# \arrow[r, "f"'] & S \end{tikzcd}$$

automorphism of untilts, then:

$$\begin{tikzcd} (S^\#)^\flat \arrow[rd, "f^\flat", "\sim"'] \arrow[d, "\sigma^\flat"'] & \\ (S^\#)^\flat \arrow[r, "f^\flat"', "\sim"] & S^\flat \end{tikzcd}$$

$\sigma^\flat = id \Longleftarrow$

$\Downarrow$ untilting

$\sigma = id$.

So if $\mathcal{X}$ presheaves of sets, $\mathcal{X}^{pfd}$ also.

2) $(-)^{pfd}$ preserves fiber products and direct limits (exercise)

3) If $\mathcal{X}$ flat sheaf, so is $\mathcal{X}^{pfd}$

4) If $\mathcal{U} \to \mathcal{X}$ formally étale

$$\mathcal{U}^{pfd} \xrightarrow{\sim} \mathcal{X}_{pfd} \times_{\mathcal{X}} \mathcal{U}$$

By def theory for $S \in \mathrm{SPfd}^{nil}$

$$\begin{tikzcd} \operatorname{Spec}(S) \arrow[r] \arrow[d] & \mathcal{U} \arrow[d] \\ \operatorname{Spf}(S^\#) \arrow[r] \arrow[ru, dashed, "\text{unique}"] & \mathcal{X} \end{tikzcd}$$

5) If R is pfd then

$$\operatorname{Spf}(R)^{pfd} \xrightarrow{\sim} \operatorname{Spf}(R)$$

Proof Let $S \in \mathrm{SPfd}^{nil}$.

$$\operatorname{Spec}(S) \to \operatorname{Spf}(R)^{pfd}$$

is

$$\begin{tikzcd} R \arrow[r] \arrow[rd] & S^\# \arrow[d, "\text{untilt}"] \arrow[ld, phantom, "\circlearrowleft"] \\ & S \end{tikzcd}$$

if this commutes then

$$(A_{inf}(R), \ker\theta) \to (W(S^\flat), I)$$

with $I = \ker\theta W(S^\flat)$

necessarily by rigidity of prims.

So there is no choice of untilt.

Also the map $R \to S^\#$ has to tilt to the map $R^\flat \to S^\flat$ so it is also uniquely determined. $\square$

6) If R is pfd then

$$\mathcal{X}^{pfd}(R) = \mathcal{X}(R)$$

Proof Have to show by def

$$\varprojlim_n \mathcal{X}^{pfd}(R/p^n) = \varprojlim_n \mathcal{X}(R/p^n)$$

But the (groupoid of) compatible collections of untilts $\{ S \xrightarrow[\text{untilt}]{} R/p^n \}_n$ consits (up to uniqe isomorphism) of $\{ R \to R/p^n \}_n$

$\square$

7) By 6) and inspection of the definition,

$\mathcal{X} \longrightarrow \mathcal{Y}$ morphism of p-adic formal

- induces $\mathcal{X}^{pfd} \xrightarrow{\sim} \mathcal{Y}^{pfd}$

$\Longleftrightarrow$

- $\forall R$ pfd $\mathcal{X}(R) \xrightarrow{\sim} \mathcal{Y}(R)$.

8) Say S is a ring s.t. there exists $S \to S_{pfd}$ an intial map to a perfectoid ring.

Then

$$\begin{tikzcd} \operatorname{Spf}(S_{pfd}) \arrow[r] \arrow[d, leftrightarrow, "\wr"] & \operatorname{Spf}(S) \\ \operatorname{Spf}(S)^{pfd} \arrow[ru] & \end{tikzcd}$$

by 7).

Proposition Let $\mathcal{X} \to \mathcal{Y}$ be finite (closed). Then $\mathcal{X}^{pfd} \to \mathcal{Y}^{pfd}$ is affine (closed).

Proof Let $S \in \mathrm{SPfd}^{nil}$. Recall

$$\operatorname{Spec}(S) \xrightarrow{\text{untilt}} \operatorname{Spf}(S^\#) \to \mathcal{Y}$$

$$\Updownarrow \text{def}$$

$$\operatorname{Spec}(S) \longrightarrow \mathcal{Y}^{pfd}$$

By alm purity (Bhatt's surj theorem)

Consider:

$$\begin{tikzcd} \mathcal{Z} \arrow[r] \arrow[d] \arrow[rd, phantom, "\lrcorner", very near start] & \mathcal{Y} \arrow[d] \arrow[r, squiggly, "(-)^{pfd}"] & \mathcal{Z}^{pfd} \arrow[r] \arrow[d, "(*)"] \arrow[rd, phantom, "\lrcorner", very near start] & \mathcal{Y}^{pfd} \arrow[d] \\ \operatorname{Spec}(S^\#) \arrow[r] & \mathcal{Y} & \operatorname{Spec}(S^\#) \arrow[r] & \mathcal{Y}^{pfd} \end{tikzcd}$$

$(*)$ is affine (closed). $\square$

Examples

1) $\mathbb{Z}_p^{pfd}(S) = \{ \text{untilts of } S \}$

2) $\underset{\sim}{\mathbb{G}_{m,R}^{pfd}}$ Fix R p-tors free pfd ring

Have

$$\begin{tikzcd} \mathbb{G}_{m,R}^{Tate} := \operatorname{Spf}(R\langle x^{\pm 1/p^\infty}\rangle) \arrow[d] \\ \mathbb{G}_{m,R} = \operatorname{Spf}(R\langle x^{\pm 1}\rangle) \end{tikzcd}$$

flat surjection of p-adic formal group schemes.

Kernel: $\mathbb{Z}_p(1)_R$; on points get

$$1 \to \mathbb{Z}_p(1)_R(S) \to S^{\flat\times} \longrightarrow S^\times \to 1$$

$$\rotatebox{90}{=}$$

$$\{ (\xi_{p^n}) \in S^{\flat\times} \mid \xi_{p^0} = 1 \}$$

$$\mathbb{Z}_p(1)_R = \operatorname{Spf}\left(R\langle x^{\pm 1/p^\infty}\rangle / (x-1) \right)$$

$\uparrow$

semiperfectoid.

$$\mathbb{Z}_p(1)_R^{pfd} = \operatorname{Spf}\left(R\langle x^{\pm 1/p^\infty}\rangle / (x-1)_{pfd} \right)$$

a "smaller subgroup scheme" of $\mathbb{G}_{m,R}^{Tate}$.

One can show

$$\mathbb{G}_m^{pfd} = \mathbb{G}_{m,R}^{Tate} / \mathbb{Z}_p(1)_R^{pfd}.$$

2) Perfectoid atlases

Theorem Let $\mathcal{X}$ p-adic formal scheme. Then $\exists$

$$\underset{\text{pfd formal scheme}}{\mathcal{U}} \xrightarrow[\text{flat cover.}]{} \mathcal{X}$$

Proof $\mathcal{X} = \mathrm{spf}(R)$ wlg.

Say $\mathcal{X} \hookrightarrow \mathcal{Y}$ s.t thm is showed for $\mathcal{Y}$. Then

$$\begin{tikzcd} \mathcal{U} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{U}' \arrow[d] \\ \mathcal{X}^{pfd} \arrow[r, hook] & \mathcal{Y}^{pfd} \end{tikzcd}$$

gives the general assertion by base change.

So the case of $\mathbb{Z}_p\langle x_i\rangle_{i\in I}$ suffices. Let

$$\begin{tikzcd} \mathcal{U} = \mathrm{Spf}\left(\mathbb{Z}_p\langle p^{1/p^\infty}, x_i^{1/p^\infty}\rangle_{i\in I}\right) \arrow[d] \\ \mathcal{X} \end{tikzcd}$$

$$= \mathbb{Z}_p\langle x_i^{1/p^\infty}\rangle[t^{1/p^\infty}]^{p\wedge} \big/ (t-p)$$

$\mathcal{U}$ is affine perfectoid formal.

Take $\mathcal{U} \to \mathcal{X}^{pfd}$ induced.

We claim that it is a flat cover; take $S \in \mathrm{SPfd}^{nil}$, $\mathrm{spec}(S) \to \mathcal{X}^{pfd}$.

This is $\mathrm{Spf}(\underset{\substack{\text{untilt of}\\ S}}{S^\#}) \to \mathcal{X}$ (*).

(*) corresponds to $s_i \in S^\#$ $i \in I$.

Consider the pullback

$$\begin{tikzcd} \mathrm{Spf}(A) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{U} \arrow[d] \\ \mathrm{Spf}(S^\#) \arrow[r] & \mathcal{X} \end{tikzcd} \quad (**)$$

, with:

$$A = S^\#\langle t^{1/p^\infty}, x_i^{1/p^\infty}\rangle_{i\in I} \big/ (t-p, x_i - s_i)$$

The proof of André's lemma exactly says that for such pfdizations

$$S^\# \longrightarrow (A)_{pfd}$$

is p-comp. faith flat.

Taking "pfd" of (**) concludes □

Corollary $\mathcal{X}$ p-adic formal scheme

Take $\mathcal{U} \to \mathcal{X}$ as above. Then

$$\mathrm{Coequ}\Big(\underset{\text{pfd formal scheme}}{\mathcal{U}\times_{\mathcal{X}^{pfd}}\mathcal{U}} \rightrightarrows \mathcal{U}\Big) = \mathcal{X}^{pfd}$$

where the coequalizer is taken in flat sheaves.

So $\mathcal{X}^{pfd}$ is a "flat algebraic space in pfd formal schemes".

3) Perfectoid Crystals

Def Perfectoid crystals :

$$\mathrm{Cris}(\overline{/A}_{\mathcal{X}}) := \mathcal{D}(\mathcal{X}^{\mathrm{pfd}})$$

where $\mathcal{X}$ is p-adic formal. ☐

Have $\mathcal{D}(\mathcal{X}^{\mathrm{pfd}}) = \varprojlim_{\substack{\mathrm{Spec}(S) \to \mathcal{X} \\ S \in \mathrm{Spfd}^{\mathrm{nil}}}} \mathcal{D}(S)$ by def.

Actually:

Prop $\mathcal{X}$ p-adic formal.

$$\mathrm{Cris}(\overline{/A}_{\mathcal{X}}) := \varprojlim_{\substack{\mathrm{Spf}(R) \to \mathcal{X} \\ R \text{ pfd}}} \widehat{\mathcal{D}}(R)$$

Proof Affine case suffices. $\mathcal{X} = \mathrm{Spf}\, R$

If $S \in \mathrm{Spfd}^{\mathrm{nil}}$

$\mathrm{Spec}(S) \longrightarrow \mathcal{X}^{\mathrm{pfd}}$ point

Corresponding to $\mathrm{Spf}(S^\#) \longrightarrow \mathcal{X}$.

$$\begin{tikzcd}
R \arrow[r] \arrow[rd] & S^\# \arrow[r] \arrow[d, "p^n \to 0"] & S^\#/p^n \arrow[ld, dashed, "\exists"] \\
& S &
\end{tikzcd}$$

for some n

The existence of such a factorization shows that

$$\{ R \to S^\#/p^n \}_{n, S^\# \text{ pfd}}$$

is final in the system of all $\mathrm{Spf}(R)^{\mathrm{pfd}}$ — points ☐

Corollary $\mathcal{X} = \mathrm{Spf}(R)$. Then

in derived rings

$$R\Gamma(\mathcal{X}^{\mathrm{pfd}}, \mathcal{O}) = \varprojlim_{\substack{(S \leftarrow R) \\ \text{pfd}}} S$$

$$= R\Gamma_{\mathrm{arc}}(\mathrm{Spf}(R), \mathcal{O})$$

as defined previously.

Proof Take Endomorphism of the unit in the above prop.

Upshot: A perfectoid crystal on $\mathcal{X}$ is a collection

$$(\mathcal{M}_{R,f})_{R \text{ pfd},\ f: \mathrm{Spf} R \to \mathcal{X}}$$

with

$$\begin{tikzcd}
\forall\ \mathrm{Spf} R' \arrow[rrd, "f'"] \arrow[dd, "g"'] & & \\
& \circlearrowleft & \mathcal{X} \\
\mathrm{Spf} R \arrow[rru, "f"'] & &
\end{tikzcd}$$

, have

$$g^* \mathcal{M}_{R,f} \simeq \mathcal{M}_{R',f'}$$

Such that identifications satisfy "higher coherences" analogous to cocycle conditions.

4) Affineness

Given R p-complete, we have defined

$$R_{pfd}$$

which is in general a non-discrete coconnective derived ring.

Using "Higher algebra" there is a notion of (p-complete) modules over R_{pfd}

$$\widehat{\mathcal{D}}(R_{pfd})$$

(up to p-completion)

Theorem Say R is $\widehat{\text{topologically}}$ of finite over some perfectoid ring.

Then if $\mathfrak{X} = \mathrm{spf}(R)$

$$\mathcal{D}(\mathfrak{X}^{pfd}) \simeq \mathcal{D}(R_{pfd})$$

Sketch of **Proof** Under this hypothesis there exists

$$\begin{tikzcd} & R \arrow[dl] \arrow[dr] & \\ R_{pfd} \arrow[rr] & & S \end{tikzcd}$$

↙ pfd ring

which induces a perfectoid atlas and $R_{pfd} \longrightarrow S$ is "descendable".

Descendable is a property of maps of derived rings (E_∞ rings) that ensures

$$\mathcal{D}(R_{pfd}) = \varprojlim_{\Delta} \mathcal{D}\left(S^{\otimes^\bullet_{R_{pfd}}}\right)$$

Where $S^{\otimes^\bullet_{R_{pfd}}}$ is the cosimplicial object

$$S \rightrightarrows S \hat{\otimes}_{R_{pfd}} S \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; S \hat{\otimes}_{R_{pfd}} S \hat{\otimes}_{R_{pfd}} S (\cdots).$$

Crucially as $(-)_{pfd}$ commutes with fiber products and that $S^{\hat{\otimes}^n_R}$ is semiperfectoid we have

$$S^{\hat{\otimes}^n_{R_{pfd}}} \cong \left(S^{\hat{\otimes}^n_R}\right)_{pfd}$$
$$\cong \mathcal{O}\left(\left(\mathrm{Spf}(S)^{\times^n_{\mathrm{Spf}(R)}}\right)^{pfd}\right)$$

Now

$$\mathcal{D}(\mathrm{spf}(R)^{pfd}) = \varprojlim_{\Delta} \mathcal{D}\left(\left(\mathrm{Spf}(S)^{\times^n_{\mathrm{Spf}(R)}}\right)^{pfd}\right)$$
$$= \varprojlim_{\Delta} \mathcal{D}\left(\left(S^{\hat{\otimes}^n_R}\right)_{pfd}\right)$$

because $\mathrm{spf}(S) \to \mathrm{spf}(R)^{pfd}$ is a pfd atlas by flat descent. $\square$

Using the same "Čech-Nerve identifications", we have:

Theorem Under the same hypothesis

$$\begin{tikzcd} \mathrm{Map}_R(\mathrm{Spf}(S), \mathfrak{X}^{pfd}) \arrow[d, "\wr"] \\ \mathrm{Der}_R(R_{pfd}, S) \end{tikzcd}$$

$\forall S$ p-complete.

↙ Calculated in ∞-cat of R-derived algebras.

↑ In particular this Hom space is in fact discrete!